

```
In [5]: import seaborn as sns
import matplotlib.pyplot as plt
```

```
In [4]: !pip install seaborn
```

```
Requirement already satisfied: seaborn in c:\users\chinmay\appdata\local\programs\python\python311\lib\site-packages (0.13.2)
Requirement already satisfied: numpy!=1.24.0,>=1.20 in c:\users\chinmay\appdata\local\programs\python\python311\lib\site-packages (from seaborn) (2.4.2)
Requirement already satisfied: pandas>=1.2 in c:\users\chinmay\appdata\local\programs\python\python311\lib\site-packages (from seaborn) (2.2.2)
Requirement already satisfied: matplotlib!=3.6.1,>=3.4 in c:\users\chinmay\appdata\local\programs\python\python311\lib\site-packages (from seaborn) (3.10.9)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\chinmay\appdata\local\programs\python\python311\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.3.3)
Requirement already satisfied: cycler>=0.10 in c:\users\chinmay\appdata\local\programs\python\python311\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\chinmay\appdata\local\programs\python\python311\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (4.62.1)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\chinmay\appdata\local\programs\python\python311\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.5.0)
Requirement already satisfied: packaging>=20.0 in c:\users\chinmay\appdata\local\programs\python\python311\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (26.2)
Requirement already satisfied: pillow>=8 in c:\users\chinmay\appdata\local\programs\python\python311\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (12.2.0)
Requirement already satisfied: pyparsing>=3 in c:\users\chinmay\appdata\local\programs\python\python311\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in c:\users\chinmay\appdata\local\programs\python\python311\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\chinmay\appdata\local\programs\python\python311\lib\site-packages (from pandas>=1.2->seaborn) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in c:\users\chinmay\appdata\local\programs\python\python311\lib\site-packages (from pandas>=1.2->seaborn) (2025.3)
Requirement already satisfied: six>=1.5 in c:\users\chinmay\appdata\local\programs\python\python311\lib\site-packages (from python-dateutil>=2.7->matplotlib!=3.6.1,>=3.4->seaborn) (1.17.0)
```

```
[notice] A new release of pip is available: 24.0 -> 26.1.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```

```
In [6]: titanic = sns.load_dataset('titanic')
```

```
In [7]: titanic
```

Out[7]:

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adl
0	0	3	male	22.0	1	0	7.2500	S	Third	man	
1	1	1	female	38.0	1	0	71.2833	C	First	woman	
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	
3	1	1	female	35.0	1	0	53.1000	S	First	woman	
4	0	3	male	35.0	0	0	8.0500	S	Third	man	
...	...	...	...	...	...	...	...	...	...	...	...
886	0	2	male	27.0	0	0	13.0000	S	Second	man	
887	1	1	female	19.0	0	0	30.0000	S	First	woman	
888	0	3	female	NaN	1	2	23.4500	S	Third	woman	
889	1	1	male	26.0	0	0	30.0000	C	First	man	
890	0	3	male	32.0	0	0	7.7500	Q	Third	man	

891 rows × 15 columns



In [8]:

titanic.describe()

Out[8]:

	survived	pclass	age	sibsp	parch	fare
count	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

In [9]:

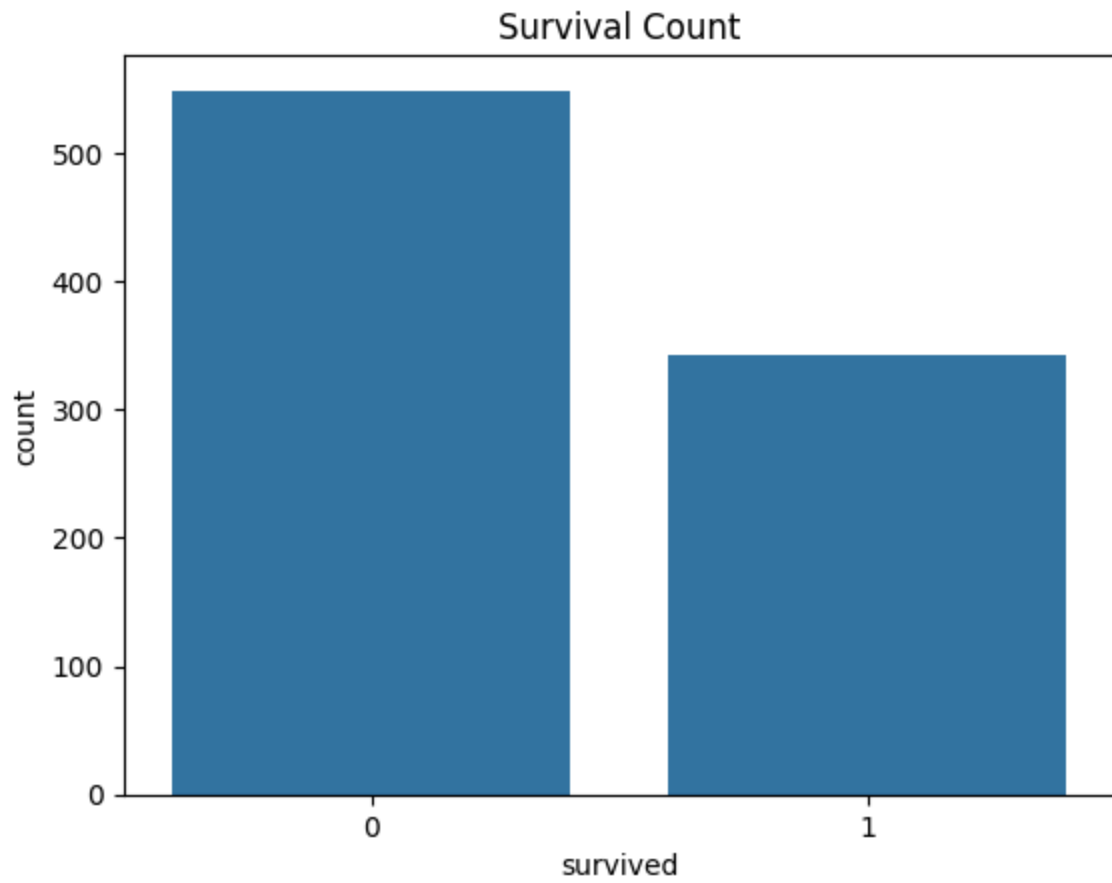
titanic.isnull().sum()

```
Out[9]: survived      0
        pclass        0
        sex           0
        age          177
        sibsp         0
        parch         0
        fare          0
        embarked      2
        class         0
        who           0
        adult_male     0
        deck          688
        embark_town    2
        alive          0
        alone          0
        dtype: int64
```

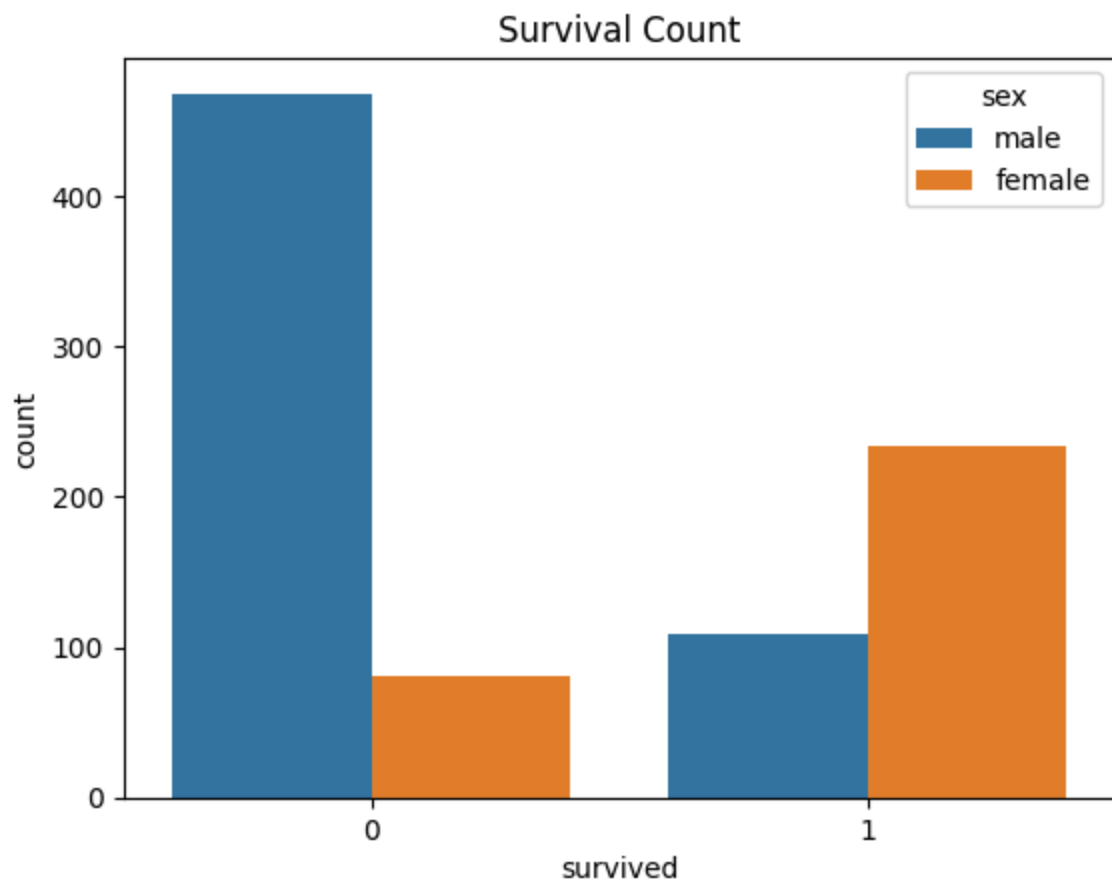
```
In [11]: titanic.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   survived    891 non-null    int64
1   pclass      891 non-null    int64
2   sex         891 non-null    object
3   age         714 non-null    float64
4   sibsp       891 non-null    int64
5   parch       891 non-null    int64
6   fare        891 non-null    float64
7   embarked    889 non-null    object
8   class       891 non-null    category
9   who         891 non-null    object
10  adult_male   891 non-null    bool
11  deck         203 non-null    category
12  embark_town  889 non-null    object
13  alive        891 non-null    object
14  alone        891 non-null    bool
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 80.7+ KB
```

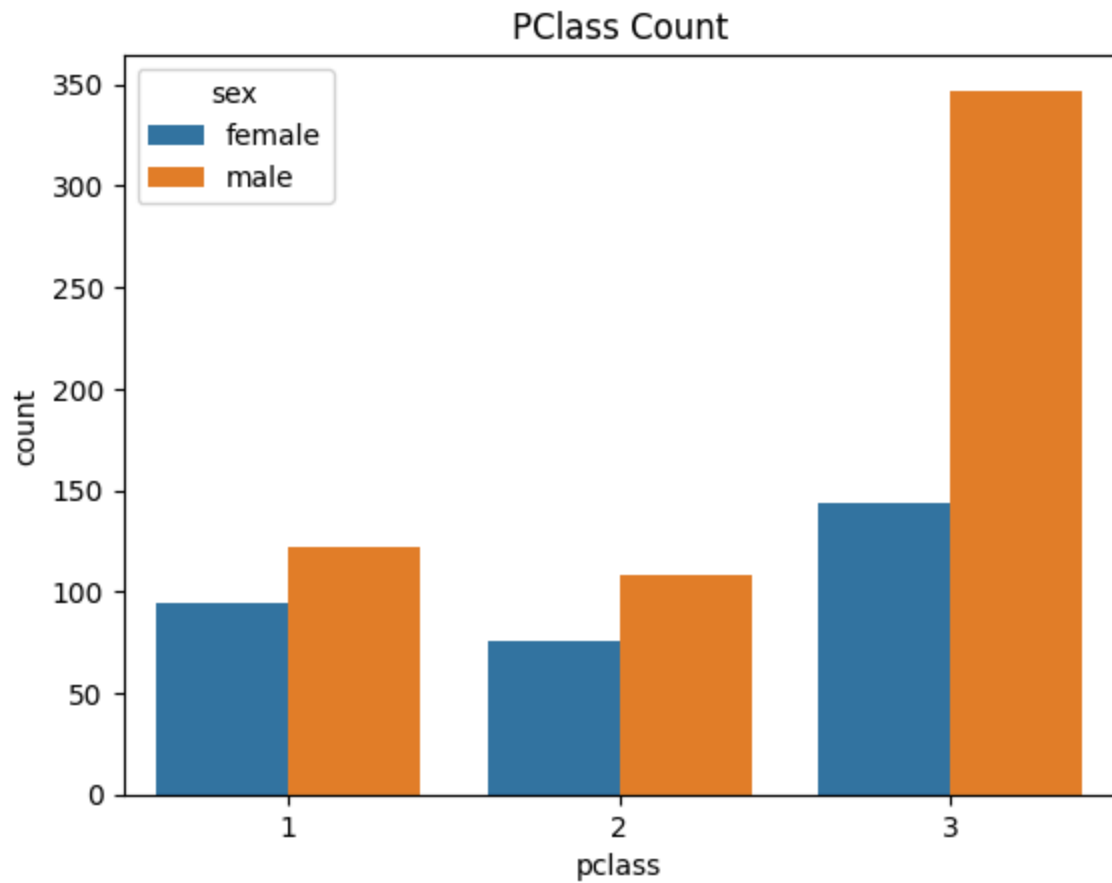
```
In [23]: sns.countplot(x='survived',data=titanic)
         plt.title('Survival Count')
         plt.show()
```



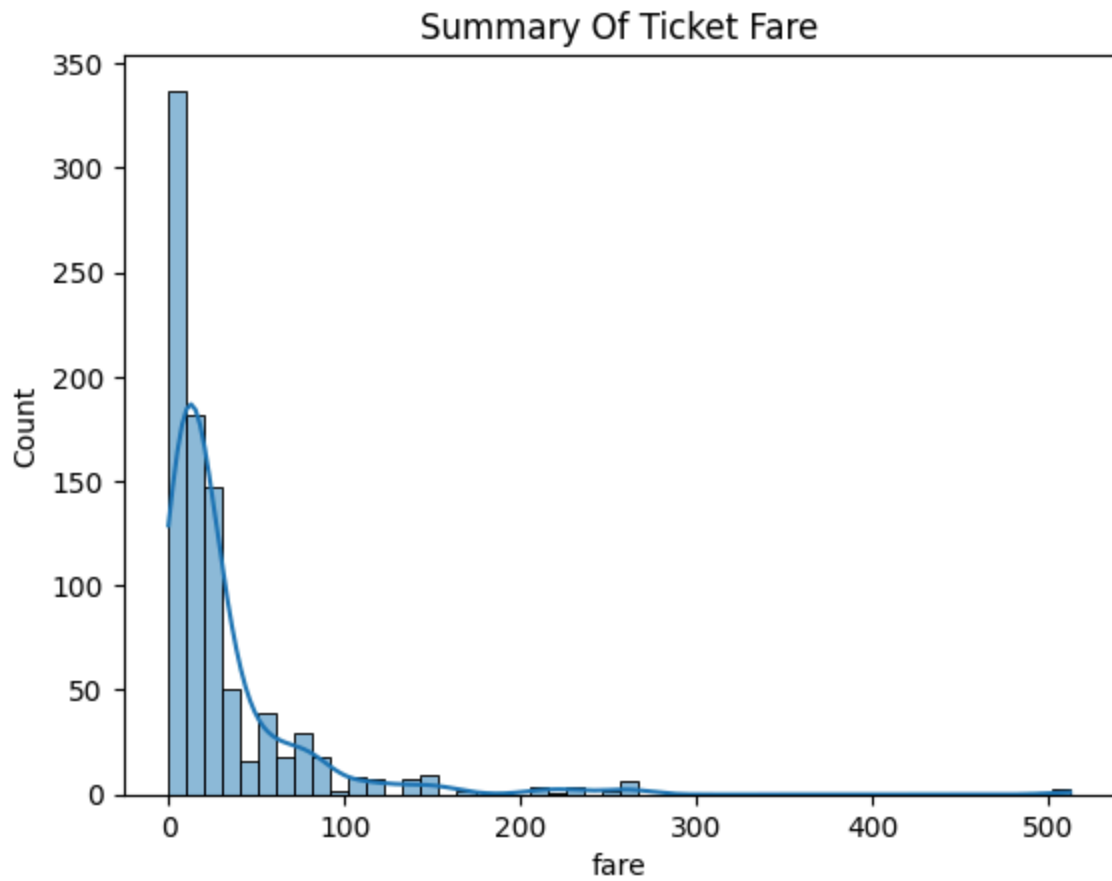
```
In [21]: sns.countplot(x='survived', hue='sex', data=titanic)
plt.title('Survival Count')
plt.show()
```



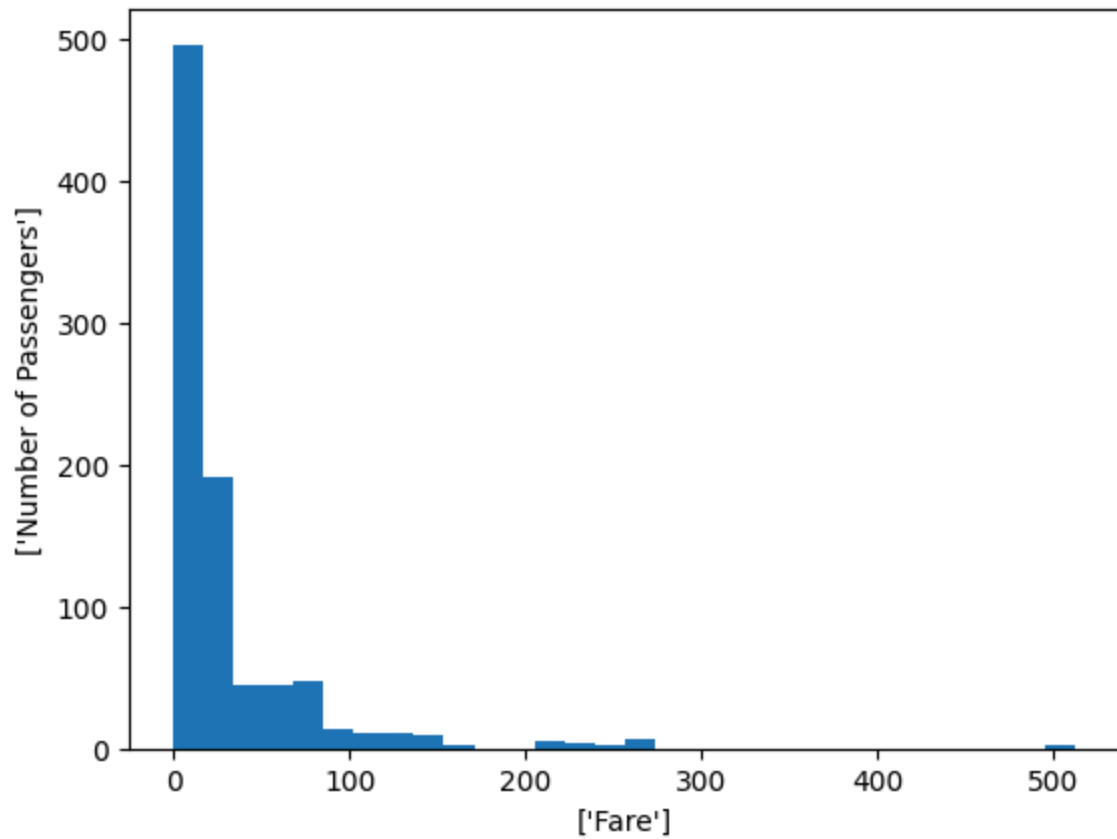
```
In [25]: sns.countplot(x='pclass',hue='sex',data=titanic)
plt.title('PClass Count')
plt.show()
```



```
In [33]: sns.histplot(titanic['fare'],bins = 50, kde=True)
plt.title('Summary Of Ticket Fare')
plt.show()
```



```
In [32]: plt.hist(titanic['fare'], bins = 30)
plt.xlabel(['Fare'])
plt.ylabel(['Number of Passengers'])
plt.show()
```



In []: